EFB 654*: Introduction to R and Reproducible Research

*EFB 496 for undergrads

1 General Information

Credits: Two
Meets: MW, 10:35-11:30am
Meeting Location: Baker 309
Instructor: Hyatt Green
Email: hgreen@esf.edu
Phone: 315-470-4814
Office: 201 Illick Hall
Office hours: W 1-2pm

2 Course Description

2.1 Purpose

Although technological advances and automation, such as next-generation sequencing and remote sensing, have greatly benefited and revolutionized research in many fields, they have also dramatically increased the size and number of typical datasets and created an enormous backlog of potentially valuable unanalyzed data. R allows us to wield such large datasets and, thus, draw conclusions from potentially useful data in many cases. At the same time, scientists can also do a better job of ensuring validity of their analysis and communicating their results. The purpose of this course is to familiarize students with R and associated tools to a) allow handling and analysis of large or complex data and b) promote good practices for reproducible research.

R, a statistical programming language and environment, offers capabilities to analyze data sets, big or small. RMarkdown, an authoring format, allows you to embed your R work (data summaries, stats, plots, etc.) within plain text. The two are linked together by `knitr` which evaluates and translates any R code or results imbedded in a RMarkdown document so it's readable by a translator (e.g. pdfLaTeX, pandoc). Separately, these programs are extremely useful, but using them together will, in the end, help you accomplish many of the following *research* objectives.

- Drastically improve reproducibility of your analyses
- Save hours, days, or weeks of your time by automating your analysis
- Improve your document's appearance which really does impress people
- Organizes your work and ideas so that they are more accessible to you or others
- Imbeds your analysis methods along side your raw data and final results so they can be evaluated as a single body of work simultaneously

These tools are also dramatically changing the way we communicate science for the better. Analyses can be more easily thoroughly criticized. Figures look better. Coauthors or colleagues don't have to depend on your honesty (or competency). They can take your data and your functions and easily demonstrate to themselves that you did things right or wrong *OR* make some of their own tweaks and show you why their way is better.

Because these tools can be difficult to learn, especially if you don't have any coding experience, this course is intended to get you over that initial hump so you can explore R's packages and RMarkdown's capabilities on your own. If you're ever discouraged and think that R is too difficult or may not be useful I want to emphasize the enormous competitive advantage you will have over someone without these skills when it comes to getting into graduate school, getting a postdoc, or even a tenure-track position.

2.2 Course Organization

We'll start learning both R and RMarkdown simultaneously. Luckily, R Studio makes the RMarkdown part fairly easy (e.g., 'Knit') so we'll primarily be learning the R language. Learning R is sort of like learning a language like French, Spanish, etc.—you lose it unless you use it. Entonces, there will be frequent in class and out of class exercises in addition to the final project. See Section 9: Evaluation for a point break-down.

3 My Background

I'm no expert in R, RMarkdown, or statistics, although I have been using R since 2006. I got into it on my own out of necessity and have probably made every mistake there is to make at least three times. Because today's world is flooded with data, I don't feel right sending students, especially ones that might stay in the sciences, out of ESF without the tools to analyze larger datasets. With the current pressing global issues, environmental scientists in particular will fall under intense scrutiny and need to have the tools to manage large, complex data sets in an effective and responsible way.

4 Your Background

You're a grad student or undergrad involved in research that needs to crunch some data for a paper or your thesis. You're thinking that the final project for this course is a great way to analyze that difficult data set you just produced. This doesn't necessarily have to be a large data set. If you don't have any data from your own research you'll need to find some data to work with soon. You might have read in a recent paper that the analysis you want to recreate was done in R with so-and-so package. You might not have a clue what R is—you just heard it might be something you need to learn. You also have a laptop and have downloaded the required programs and packages for this course. You have lots of time and endless determination to learn R. If you already use R regularly you will find this course a light refresher, but mostly boring. If this is you, consider giving up your seat for another student that has less experience with R—this course is in high demand.

5 Skill Outcomes

5.1 R

- Get data into and out of R
- Obtain data summaries
- Plot basic data summary plots
- Perform basic built-in statistical functions in R
- Understand basic model fitting and interpreting results
- Write unique R functions efficiently
- Know how R handles special data classes, like dates
- Understand that there are many highly specialized packages available through R
- Reach beyond course content to individualize R to your needs

5.2 Reproducible Research

- Learn basic RMarkdown
- Demonstrate literate programming

6 Prerequisites

There are no course prerequisites. Basic stats may be helpful at some points. Programming experience (Perl, Python, Matlab, C++, etc.) would be extremely helpful, but certainly not required since this course is designed to start at square one.

7 Required Materials

A laptop. **Please attend the first lecture with the latest R and RStudio installed.** Also, install the required packages for R. Confirm proper install and that things are working properly before the first class.

7.1 Is LaTeX required?

No. LaTeX is a type setting language that enables you to produce good-looking documents (PDFs) with R code and output embedded. I highly recommend you at least become familiar with LaTeX; however, its installation takes up a relatively large amount of memory (maybe 4-5 G or so). You will need LaTeX if you choose to make your final presentation using beamer.

8 Use of AI

Use all the AI you need. It's great at generating code that's error free, good at explaining how that code works, but not great experimental design, bringing meaning to results, and a few other important steps where humans are useful, for the time being. I would check out RTutor if you haven't already (http://rtutor.ai).

9 Evaluation

9.1 Participation 50%

I expect active participation in all exercises and discussion. I may take attendance at some points, but mostly I'll be gauging student participation as we go.

9.2 Final Project 50%

Your final project will consist of two components: the final written report and the final verbal presentation. You should start with data you produced, if possible. Ideally, I'd like your analysis through this course to end up in your thesis, or some other meaningful place. If you don't have any data of your own to work on, contact me and I might be able to help. **DO NOT** leave this project till the last minute. Start thinking about it today.

Near the beginning of the term we'll meet privately to discuss a dataset you've chosen to make sure it meets the course criteria. At this initial meeting, bring your first and second choices for a final project dataset. **If you don't have your own data, check out the resources in Section 16 of this syllabus**. Keep in mind you'll have to produce some real results from the data by the end of the term, so don't pick a dataset that's too large or complex. On the other hand, if you pick a dataset that's too simple (e.g., a 2×10 data frame), you likely won't be able to demonstrate R proficiency. Important dates for the final project are listed in Table 1.

Table 1: Important dates for the final project.

Deadline	Purpose	Materials Due
Feb 12	Initial data meeting	1-2 datasets
April 9-28	Final Presentations	12-15 minute presentation
May 6	Final Project Due	Final Report

9.2.1 Final Written Report 30%

The final report is the culmination of all your work in the course. In this document, you'll demonstrate most if not all of the course learning outcomes. It should be constructed using R and knitr and describe in clear detail the background, data collection, analysis methods, results, discussion, and conclusions of your chosen project. You will be evaluated according to the grading rubric (Table 4).

Submit your final project with 1) your (knitted) **pdf** file (preferred, but HTML acceptable), 2) any data in an ".Rdata" file that your source document requires, and 3) your source document ".Rmd" that you used to create your final document.

Table 2: Penalties for late work.

Hours Overdue	Penalty
24	20%
48	50%
>48	100% (no credit given)

9.2.2 Final Verbal Presentation 20%

Your verbal communication, more so than your content, will be evaluated through your final presentation. I prefer you use RMarkdown (through beamer, slidy, or IOslides) for your final presentation, but use PowerPoint if you must. Your presentation will be evaluated according to the grading rubric (Table 5). Presentations should be about **about 9 minutes leaving about 3 minutes for questions/discussion for a max of about 12 minutes total per presentation**. My hope is that these presentations will be given in-person.

Because we have to fit everybody in the allocated presentation days, make sure you are not delayed by technical difficulties. If you are first to present, please try to show up a few minutes early to set up your presentation.

You will not turn in your presentation for a grade.

It's possible undergrads will not have the opportunity to present due to time constraints. In this case, the final written report will be worth 50% of your grade.

Penalties for late work will be assessed according to Table 2.

	Date	Content	Packages	Items Due
Basic R	Jan 13	Orientation, R, CRAN, getting help		
	Jan 15	R operators, built-in functions and data sets, and saving your work		
	Jan 20	*«No Class»* MLK Jr Day		
	Jan 22	Data Classes		
	Jan 27	Data Structures		
	Jan 29	Indexing vectors		
	Feb 3	Indexing matrices, data frames, & lists	vegan	
	Feb 5	Importing/exporting data		
	Feb 10	Source doc formatting; sorting/subsetting data	picante, rmarkdown	
	Feb 12	Plotting basics I, `par()`, R colors	RColorBrewer, car, ggplot2	Data Meeting
	Feb 17	*«No Class»*		
	Feb 19	*«No Class»*		
	Feb 24	Plotting basics II		
	Feb 26	Summarizing data (`table()`, etc.); MWEs	Lahman, Hmisc	
	Mar 3	Aggregating data		
	Mar 5	Merging data		
	Mar 9-16	☼*«Spring Break»* ☼		
Intermediate R	Mar 17	ddply and reshaping data	car	
	Mar 19	Conditionals, loops		
	Mar 24	Basic stats functions, linear regression	plyr, MASS	
	Mar 26	Dates		
	Mar 31	Plotting using ggplot2		
	Apr 2	Chaining with %>% ('pipe')	magrittr, dplyr	
	Apr 7	Writing/debugging functions		
Student Pres.	Apr 9	Student presentations I		
	Apr 14	Student presentations II		
	Apr 16	Student presentations III		
	Apr 21	Student presentations IV		
	Apr 23	Student presentations V		
	Apr 28	Student presentations VI		
	May 6	*«No Class»*		Final Report

Table 3: Course schedule

	POOR	OK	EXCELLENT
Background	Little or no description of the scientific question(s) at hand	Some description of scientific questions, but no references	Thorough introduction with references
Data Collection	No description of data collection methods	Some description of data collection methods	Contains enough information about data collection to be reproduced accurately by an outside group
Data Summary	Contains no data summary	Contains data summary	Contains a thorough data summary highlighting data dimensions, meaning of particular data entries, reason for missing data if any, etc.
Analysis Methods	Uses only basic R methods. May not be descriptive enough to be reproducible. Uses R to generate results	Uses functions from specialized packages Uses R to generate results.	Uses package and custom functions. Appears to be fully reproducible. Uses efficient R coding to generate results.
Results	An incomplete description of results in the text. No figures or tables.	Good description of results. May contain tables or figures, but they may not be optimal for showing key findings or poorly formatted.	Thorough description of Results. No presentation of redundant data. Publication quality figures. Inline reporting of some Results.
Discussion	Insufficient discussion of the meaning of these results within the broader scope.	Some discussion of broader implications and/caveats of the analysis	Thorough discussion of broader implications, caveats or weaknesses of the analysis, suggested alternatives to analysis, and future experiments in this area.
Conclusions	No conclusions based on the data.	Inaccurate conclusions based on the data.	Accurate conclusions based on the data.
Formatting	Unorganized. Poor grammar or spelling. Lines may run off the page. Missing essential document elements like a table of contents or other essential information.	Somewhat organized. Has all the essential document elements.	Contains essential document elements. Uses necessary packages for proper formatting (e.g. xtable). Correct spelling and grammar throughout.
R programming	Very little or no R programming. Inefficient use of code.	Uses R throughout but could have been much more efficient.	Uses R throughout in the most efficient ways without unnecessary lines of code.

Table 4: Grading Rubric for Final Report

	POOR	OK	EXCELLENT
Project Description	Background insufficient to setup project.	Some background, but too little or too much detail.	Complete but concise background to setup the scientific question.
Experimental Design	Insufficient explanation.	Some explanation.	Sufficient but concise explanation.
Data Description	No data summary provided.	Too little or redundant data summary provided.	Thorough data summary highlighting data dimensions, meaning of particular data entries, reason for missing data if any, etc.
Analysis Methods	No description of the methods that were used.	Some description of the methods, but missing some key points important for understanding the results.	Description of any custom functions, package functions, important object classes, etc.
Results	Poor or lacking figures and tables.	Some figures and/or tables, but may be unclear or redundant.	Perfect figures and tables that are used effectively.
Conclusions	No conclusions based on the data.	Inaccurate conclusions based on the data.	Accurate conclusions based on the data.
Formatting	Content runs off slides. Unaligned or uncentered content, e.g..	Content completely on slides. All text readable.	Flawless formatting all around. Uses lists or bullets. Tables properly formatted.

Table 5: Grading Rubric for Final Presentation

10 Students with Learning and Physical Disabilities

SUNY-ESF works with the Office of Disability Services (ODS) at Syracuse University, who is responsible for coordinating disability-related accommodations. Students can contact ODS at 804 University Avenue- Room 309, 315-443-4498 to schedule an appointment and discuss their needs and the process for requesting accommodations. Students may also contact the ESF Office of Student Affairs, 110 Bray Hall, 315-470-6660 for assistance with the process. To learn more about ODS, visit `http://disabilityservices.syr.edu`. Authorized accommodation forms must be in the instructor's possession one week prior to any anticipated accommodation. Since accommodations may require early planning and generally are not provided retroactively, please contact ODS as soon as possible.

11 Academic Dishonesty Statement

Academic dishonesty in any form will not be tolerated. This includes, but is not limited to, turning in work that is not your own in its entirety, using unauthorized materials during tests or quizzes, aiding others in dishonest behavior, or any attempts to deceive the instructor about anything. Any participation in academic dishonesty will result in a failing grade. Checks for plagarism from your peers and internet resources will be performed routinely. I'm actually pretty proud of my plagarism checker R scripts.

Academic dishonesty is a sign of disrespect and breach of trust between a student, one's fellow students, or the instructor(s). By registering for courses at ESF you acknowledge your awareness of the ESF Code of Student Conduct (`http://www.esf.edu/students/handbook/StudentHB.05.pdf`), in particular academic dishonesty includes but is not limited to plagiarism and cheating, and other forms of academic misconduct. The Academic Integrity Handbook contains further information and guidance (`http://www.esf.edu/students/integrity/`). Infractions of the academic integrity code may lead to academic penalties as per the ESF Grading Policy (`http://www.esf.edu/provost/policies/documents/GradingPolicy.11.12.2013.pdf`).

12 Downloading R and RStudio

To download R visit: `http://www.r-project.org/`.
To download R Studio visit: `http://www.rstudio.org`.

13 Getting Started with R and RStudio

13.1 Windows

On the Windows download page, click on base. Download the installer, Download the latest stable verison of R for Windows. It will take a few minutes to download, and I recommend you save it to your desktop (instead of running it directly from the web). Once the download is complete, double-click the icon to start the installation. Simple questions are posed during the installation. Accept the default settings.

13.2 Mac OS X

The R Project page provides some helpful advice depending on your version of the Mac OS. Follow the instructions, and if there is a problem we can help you figure it out.

14 Getting Started with LaTeX

14.1 Downloading LaTeX

To download LaTeX visit: `http://\LaTeX-project.org/ftp.html`

15 Download/Install Help

Do not contact me for help downloading and installing R or RStudio. I say this not because I'm an unhelpful SOB, but because you need to be able to figure this out on your own. But really, if you've tried all you can think of and then tried three more times and you're still SOL, contact me. I'll be practically useless when it comes to trouble-shooting LaTeX installs. Although I've found LaTeX extremely useful in the past 11 years or so, it is not my forte. Fortunately, most of the time LaTeX downloads and installs integrate seamlessly with R Studio. We'll have to work together on solving any issues with LaTeX .

16 Other Useful Resources

16.1 Books

If you really want to get into it:
The Structure and Interpretation of Computer Programs by Harold Abelson and Gerald Jay Sussman

Advanced R by Hadley Wickham
See: `http://www.r-project.org/doc/bib/R-books.html`

R for Data Science by Hadley Wickham
See: `https://r4ds.had.co.nz/index.html`

Handling and Processing Strings in R by Gaston Sanchez
See: `http://gastonsanchez.com/Handling_and_Processing_Strings_in_R.pdf`

16.2 Blogs/Websites

16.2.1 R Resources

CRAN: `http://r-project.org`
TaskViews: `http://cran.r-project.org/web/views`
Advanced R: `http://adv-r.had.co.nz`
Quick-R: `http://www.statmethods.net`
R-bloggers: `http://www.r-bloggers.com`
RTutor: `http://rtutor.ai`

16.2.2 Authoring Formats

R Markdown v2: `http://rmarkdown.rstudio.com`
Beamer Theme Matrix: `https://www.hartwork.org/beamer-theme-matrix/`
Slidify tutorial: `https://www.uvm.edu/rsenr/vtcfwru/R/fledglings/14_Slideshows.html`
Another Slidify tutorial: `http://slidify.github.io/workshops/tutorials/03/#21`

16.2.3 Minimal Working Examples

Minimal Working Examples (general): `http://stackoverflow.com/help/mcve`
Minimal Working Examples (from R-bloggers): `http://www.r-bloggers.com/minimal-reproducible-examples/`
Minimal Working Examples (from StackOverflow): `http://stackoverflow.com/questions/5963269/how-to-make-a-great-r-reproducible-example`

16.2.4 Spatial Analysis

spatial cheatsheet: `http://www.maths.lancs.ac.uk/~rowlings/Teaching/UseR2012/cheatsheet.html`
spatial tutorial: `https://github.com/Robinlovelace/Creating-maps-in-R`

16.2.5 Miscellaneous

Another R course for Python Programmers: `https://ramnathv.github.io/pycon2014-r/`
Debugging in R: `http://www.stats.uwo.ca/faculty/murdoch/software/debuggingR/`

qqplot2 quick reference: `http://zevross.com/blog/2014/08/04/beautiful-plotting-in-r-a-ggplot2-cheatsheet-3`
help on reshaping data: `http://seananderson.ca/2013/10/19/reshape.html`
Ordination for Ecologists: `http://ordination.okstate.edu`
Writing formulae: `http://ww2.coastal.edu/kingw/statistics/R-tutorials/formulae.html`
R popularity: `http://r4stats.com/2015/10/13/rexer-analytics-survey-results/`
Networks in R: `http://kateto.net/network-visualization`

16.3 Data

Inside-R post: `http://www.inside-r.org/howto/finding-data-internet`
rOpenSci: Packages to get data: `http://ropensci.org/packages/`